26900405 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCPHY300 - CLASSICAL MECHANICS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **3** marks

1.	Find the Lagrangian for the motion of a bob in a simple pendulum.	[U] / [CO1]
2.	Mention any two advantages of using generalized coordinates in analytical mechanics.	[U] / [CO1]
3.	Differentiate between actual displacement and virtual displacement.	[U] / [CO1]
4.	Obtain the Lagrange's equation for angular motion in a central force problem.	[An] / [CO2]
5.	Write the expression for the transverse acceleration (a_θ) of a particle in plane polar coordinates and state its physical significance.	[An] / [CO2]
6.	Write down the effective Lagrangian of a two body problem and explain the terms involved.	[U] / [CO2]
7.	Express the action integral for a particle of mass m , moving with velocity v in a conservative potential V, in terms of its kinetic and potential energies.	[U] / [CO3]
8.	Write down Hamilton's canonical equations of motion for a conservative, holonomic system.	[K] / [CO3]

9. Write down the Hamiltonian of a planet moving under an inverse-square force field in plane polar coordinates. [A] / [CO3]

10. What was the ether hypothesis? [U] / [CO4]

11. Explain simulataneity of events as a consequence of Lorentz transformation. [A] / [CO4]

12. Differentiate between the special theory of relativity and the general theory of relativity. [K] / [CO4]

[3x10 = 30]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **8** marks

13. What are constraints? Explain the various type of constraints. Consider a pendulum with an inextensible string of length *l*. Identify the constraint and type of constraint involved. What variation in constraint is observed when string becomes extensible? [U] / [CO1]

14. Prove that the total angular momentum of the system is a constant using the idea of isotropy of space. [U] / [CO1]

15. Deduce Kepler's Second Law from the conservation of angular momentum [An] / [CO2]

16. Using the effective-potential graph, analyze the physical meaning of the cases E>0, E=0, 0>E>$(V_{eff})_{min}$, E=$(V_{eff})_{min}$, and E<$(V_{eff})_{min}$. [An] / [CO2]

17. Explain the Lagrangian and Hamiltonian approaches to classical mechanics and compare their fundamental variables and governing equations [An] / [CO3]

18. A system with two generalized coordinates x and y is described by the Lagrangian $L = a\dot{x}^2 + b\dot{y}^2 - cx^2$, where a, b and c are constants. Obtain the Hamiltonian and Hamilton's equations of motion. Identify the cyclic coordinate. [A] / [CO3]

19. Compare the Galilean and Lorentz transformations. Discuss their physical significance and explain the low-velocity limit. [U] / [CO4]

20. (a) Obtain the relation connecting the proper length and the contracted length of a moving object.
(b) Calculate the length of a scale moving with a speed of 0.8c, if its proper length is 1m. [U] / [CO4]

[8x5 = 40]